§5.2 Strong Induction and the well ordering principle

Two Techniques / Two problems

Fundamental Theorem of Arithmetic

Any integer $n \geq 2$ can be written as a product of 1 or more prime numbers, (If the factors are sorted (ascending/descnding/alphabetical/etc.) then the factorization is unique)

We'll prove the white part (The uniqueness requires chapter 4)

The natural choice for predicate is $P(n)$ is "n can be factored as a product of primes"

$P(n)$ is not linked appropriately, by implication so either:

- Modify our predicate, or
- use a different version of Induction

$P(2)$ is true because $2 = 2$ (A product of 1 prime)

$P(3)$ is true because $3 = 3$ (Also a product of 1 prime)

$P(4)$ is true because $4 = 2 \times 2$ (A product of 2 primes)

Case analysis

To write n as a product of primes:

- either
  - n is prime an we write $n = n$
  - n is composite so n has some factor d with $2 \leq d \leq n - 1$

Also

$\frac{n}{d}$ is an integer with $2 \leq \frac{n}{d} \leq n - 1$

If we have a list showing we can factor all the numbers all the numbers from 2 to $n - 1$, then

$d = p_1 \times p_2 \times ... \times p_n$ for some primes and some k

and

$\frac{n}{d} = q_1 \times q_2 \times ... \times q_n$ for some primes and some l

so

$n = l \times \frac{n}{d} = p_1 \times p_2 \times ... \times p_k \times q_1 \times q_2 \times ...q_l$

We only actually need $P(d)$ and $P\left(\frac{n}{d}\right)$, we want the whole list for simplicity

The arguement shows:

$\forall k \geq 2 P(2) \wedge P(3) \wedge ... \wedge P(k) \rightarrow P(k+1)$

Stronger Hypothesis than $P(k)$

Also

$P(2)$

Strong Induction

We have the rule

$P(l) \wedge P(l+1) \wedge ... \wedge P(l+r)$

$\forall k \geq (\text{something bigger than } l), P(l) \wedge P(l+1) \wedge ...P(l+k) \rightarrow P(l+r+1)$

$\therefore \forall n \geq l P(n)$

Why it works

Let $Q(n)$ be $P(l) \wedge P(l+1) \wedge ... \wedge P(n)$

Then as long as t is big enough, we can show $Q(t)$

and

$\forall k \geq n Q(k) \rightarrow Q(k+1)$

We could use normal induction on Q

In our example

$Q(n)$ is “we can facor $2, 3, 4..., n$”

The Well-Ordering principle

Another problem and another technique.

The Division Algorithm states:

$\forall a \in \mathbb{Z} \wedge \forall d \in \mathbb{Z}^+$

there exissts a unique $q \in \mathbb{Z}$ and $r \in \mathbb{Z}$ such that,

$a = dq + r$ and $O \leq r < d$

$q =$ quotient

$r =$ remainder

The right tool is well-ordering

Definition

The well ordering principle states:

Every non-empty subset of non-negative integers has a smallest element. (Smallest also often called least)

To find r and then q,

Let

$S = \{n \in \mathbb{N} \mid n = a - dq\}(\text{and } q \in \mathbb{Z})$

Claim S is non-negative,

If $a \geq 0$, then $a \in S$

If $a < 0$, then $a - d(a + 1) = a(1 - d - 1) = ad \in S$

So S must have a smallest element,

Let $r = \min(S)$

so

$r = a - dq_0$ for some q

Suprisingly $r < d$

Assume otherwise, so $r > d$ but then

$a - dq_0 \geq d$

so

$a - d(q_0 + 1) \geq d - d = 0$

and

$a - d(q_0 + 1) \in S$

Then $q = q_0$ completes the equation

$a = dq_0 + r$

and $0 \leq r < d$,

Wel ordering is effectively using induction in a proof by contradiction setting.

§5.3 Recursive Definitions & Structural Induction

(Structural Induction is a way to prove things about recursively defined objects.)

By example

Strings over an alphabet

Over the alphabet Σ (Where its just some set, possibly infinite)

The set of all strings:

Σ^* (Where $*$ represents everything under something, this case of strings)

Notionally

$\Sigma^* = U_{k=0}^{\infty}\Sigma^k$ where Σ^k is all $k-$ letter strings,

A different approach, suitable for proof

There is a special string λ with lenght of 0 (λ is not a letter from our alphabet)

Every other string has a last letter which comes after a prefix, which is itself a string.

Definition

Σ^* is defined by

1. $\lambda \in \Sigma^*$
2. IF $w \in \Sigma^*$ and $x \in \Sigma$, then $wx \in \Sigma^*$

In practice,

if $\Sigma = \{a, b, c, d, t\}$

then

$\lambda \in \Sigma^*$

so

$\lambda c = c \in \Sigma^*$

so

$\mathrm{ca} \in \Sigma^*$

so

$\mathrm{cat} \in \Sigma^*$